ChatGPT

Reclassifying DeFi: An Evidence-Driven Audit of DeFiFormal's Twelve-Area Taxonomy

Executive decision and research status

[Recommendation; high confidence] The current twelve-area DeFiFormal taxonomy should **not** be preserved as a flat, mutually exclusive classification of "protocols." Its central defect is not the number twelve. It is that the twelve labels classify different kinds of things at different ontological levels: financial products, financial instruments, implementation mechanisms, execution architectures, settlement infrastructure, asset provenance, and legal/custodial arrangements are placed beside one another as if they were peers. Current institutional, academic, protocol, and analytics evidence all point toward separating those levels. BIS work explicitly distinguishes blockchain, smart-contract, protocol, and application layers; a recent DeFi systematization independently separates a protocol's value proposition from its algorithm/design and network architecture; and current analytics providers use overlapping sector concepts rather than a single stable architectural partition. [1]

[Recommendation] The best replacement is **M4+**: start from candidate **M4, linked product and implementation classifications**, but make both multi-label as in M3 and attach the formal-behavior profile of M5. In practical terms, DeFiFormal should maintain four linked objects:

1. **economic-product/function taxonomy**, applied to applications, products, or markets;
2. **financial-instrument/payoff taxonomy**, applied to claims and positions;
3. **mechanism taxonomy**, applied to implementation mechanisms and transaction protocols; and
4. **infrastructure, control, and trust facets**, applied to dependencies and off-chain arrangements.

A fifth, derived **formal-behavior profile** should describe actors, state variables, transitions, obligations, conservation laws, liveness assumptions, trust boundaries, and failure modes. Protocol organizations should generally be **containers**, not atomic classified units.

This recommendation is consistent with the historical literature. BIS characterized the 2021 DeFi ecosystem mainly in terms of trading, lending and investing protocols plus stablecoins; a 2023 review described exchanges, lending, derivatives and insurance; by 2024–2026, liquid staking, restaking, risk curation, capital allocation and numerous specialized architectures had become salient enough to receive their own research or analytics treatment. That evolution is evidence that a flat list of contemporary market sectors is temporally brittle, whereas economic functions, payoff forms and trust mechanisms are much more stable abstractions. [2]

[Contradiction; very high confidence] The mandatory DeFiFormal corpus could not be reproduced. On September 2, 2026, the exact public repository path specified in the assignment returned GitHub 404. GitHub's repository filter for the `CharlesHoskinson` account returned zero repositories matching `defi`, and GitHub global repository search returned zero repositories named or matching `defiformal`. This establishes only that the requested repository was **not publicly retrievable at the research date**; it does not prove that it never existed, was not renamed, was not deleted, or is not now private. [3]

That prevents an honest completion of several corpus-dependent deliverables. I therefore do **not** invent a branch, commit, 72-protocol roster, 60-construction roster, 58-element decomposition, obligation graph, clustering result, or protocol-by-protocol old-to-new mapping. The exact requested branch and commit are consequently **unresolved, not "main/latest."**

[Methodological limitation] Scrapling was not exposed as an executable tool in this research environment. The online evidence was acquired through the provided web-acquisition system instead. Accordingly, this report cannot honestly claim the assignment's requested Scrapling provenance guarantees: immutable raw-response preservation, exact UTC acquisition time for every object, or locally computed SHA-256 for every raw page. Web observations were made during this research session on September 2, 2026; publication/update dates are given where sources supplied them. This is a material deviation from the requested acquisition protocol and should be corrected before treating the evidence collection as an archival wiki-LLM corpus.

The machine-readable portion that can be supplied without fabricating the missing corpus is available here:

Download the draft linked ontology

Download the twelve-area old-to-new crosswalk

Download the decision scorecard

Corpus reconstruction and evidence audit

[Repository observation] No exact DeFiFormal branch or commit can be recorded, because no public repository object at the supplied path could be resolved. The account itself remains publicly visible on GitHub, but both owner-scoped and global repository searches fail to surface DeFiFormal. [4]

That changes the evidentiary status of every quantity asserted in the assignment:

Corpus question	Result on September 2, 2026	Evidentiary status
Which protocols constitute the "72"?	**Not reproducible.** No committed roster was retrievable.	**[Unresolved question]**
Which 60 received application constructions?	**Not reproducible.** Expansion/construction artifacts were not retrievable.	**[Unresolved question]**
Why were the remaining 12 excluded?	Cannot be determined without selection notes/verdicts.	**[Unresolved question]**

Corpus question	Result on September 2, 2026	Evidentiary status
Does every protocol have exactly one category?	Cannot be established. The proposed ontology should not assume this.	**[Unresolved question / Recommendation]**
Did membership change between files/stages?	Cannot be compared.	**[Unresolved question]**
Did “72” ever denote mechanisms, elements, obligations, or another unit?	No retrievable source allows the unit to be verified.	**[Unresolved question]**
Is the stated “58-element vocabulary” a committed artifact?	Not independently reproduced.	**[Unresolved question]**
Can E01–E05 mechanism/ obligation experiments be reproduced?	No, because their source matrices are absent.	**[Contradiction with required experiment]**
Can an exact old-to-new mapping of all 72 be issued?	No; doing so would fabricate identities and memberships.	**[Unresolved question]**

[Inference; high confidence] The inability to reconstruct the corpus is not a minor documentation problem. The requested E02–E04 clustering experiments require an exact protocol-by-element and protocol-by-obligation matrix. Without the committed rows, feature definitions, version dates and residue rules, a clustering result cannot be reproduced even if one happened to know the names of 72 protocols. In particular, the unit “protocol organization” is insufficient: current primary documentation shows that one organization can expose materially different products and versions. Morpho Vaults introduce curator, allocator and sentinel roles and risk caps distinct from the underlying credit markets, while its documented risk behavior varies by vault/version; Sky now exposes stablecoin issuance, savings, lending and curated Morpho-based vault products from the same ecosystem. [5]

[Source fact] This is also consistent with the BIS layer model: protocols combine smart contracts into use cases, while applications sit above those protocols. Treating “organization,” “protocol,” “application” and “financial instrument” as interchangeable therefore loses information even before any DeFi-specific taxonomy is chosen. [6]

Evidence inventory. The highest-weight sources used for the part of the audit that *is* reproducible are:

Evidence	Type	Unit	Independence / confidence
GitHub exact path, owner search, global repository search	Empirical repository observation, observed 2026-09-02	Repository	Directly observed; **very high confidence** for current public availability only. [3]

Evidence	Type	Unit	Independence / confidence
DefiLlama category inventory and TVL methodology	Current empirical/ descriptive analytics	Protocol/ category/ metric	Primary analytics source; moving data; **high confidence** as an observation, not as ontology. [7]
Messari classification methodology	Descriptive analytics taxonomy	Protocol sector	Current primary provider methodology; **high confidence**. [8]
Token Terminal sector methodology	Descriptive/empirical analytics	Market sector/ metric	Primary provider; metrics intentionally sector-dependent; **high confidence**. [9]
BIS 2025 DeFi functional/ layer framework	Institutional analytical	System layers/ functions	Independent institutional evidence; **high confidence**. [6]
FSB tokenisation report	Institutional analytical/normative	Token/claim/ risk	Independent institutional evidence; **high confidence**. [10]
2024 DeFi SoK	Academic systematic taxonomy/risk work	Protocol, algorithm, architecture	Independent research; **high confidence for framework**. [11]
Morpho, Across, LayerZero, Lido, EigenLayer, Hyperliquid, Polymarket, Sky, Circle, Ethena, Ondo documentation	Normative/ descriptive primary protocol evidence	Product/ version/ mechanism	Primary documentation; promotional incentives vary; **high confidence for stated architecture**. [12]
Midnight Compact/ZKIR documentation	Normative implementation documentation	DSL/compiler/ proof artifacts	Primary documentation; **very high confidence** for current compilation flow. [13]
Marlowe implementation/ test documentation	Formal/experimental implementation evidence	DSL semantics/ deployment	Primary repository evidence; **high confidence**. [14]

The central **contradiction register** is therefore unusual: the strongest contradiction is not between two descriptions of a protocol, but between the assignment's presupposition of a publicly inspectable committed DeFiFormal corpus and the current observable GitHub state. The proper wiki-LLM treatment is to preserve that contradiction rather than fill the missing record from memory or inference. [15]

Structural audit of the twelve areas

Because DeFiFormal's own definitions were unavailable, the following are **operational reconstructions**, not claims about what `atlas.tex` or the missing lane files actually said.

Existing area	Defensible operational rule	Principal defect	Recommended treatment
D01 Spot exchange and AMMs	Classify an application/market whose principal operation is immediate exchange of on-chain assets. Exclude derivatives, routing-only services and cross-domain transfer.	"Spot exchange" is a product; **AMM is a pricing/liquidity mechanism**. Order-book or RFQ spot markets belong to the same product family.	**Rename/narrow:** Exchange & market access + `spot` instrument + AMM/ order-book/RFQ mechanism.
D02 Lending and credit	Include applications creating a repayable principal obligation between capital suppliers and users, normally with interest/ fees and possibly collateral.	Relatively coherent, but overlaps D03 whenever a stablecoin is borrowed into existence.	**Preserve/rename:** Credit & financing.
D03 CDP/ collateral stablecoins	Include systems in which collateral supports issuance of a peg-targeting protocol liability and a debt position is tracked.	Simultaneously a credit product, monetary instrument, issuance mechanism, collateral model and liquidation system. Sky's liquidations explicitly transfer undercollateralized vault debt and sell collateral. [16]	**Merge/convert:** Credit + monetary claim + collateralized-debt mechanism.
D04 Liquid staking/ restaking	Include capital delegated to consensus/security services where rewards and possible penalties accrue, with liquid claims where applicable.	LST issuance and restaking/ security leasing have materially different obligations. Lido's stETH accounting reflects rewards/ slashing and queued withdrawal; EigenLayer lets AVSs define slash conditions over delegated stake. [17]	**Split under a common staking/ security parent.**

Existing area	Defensible operational rule	Principal defect	Recommended treatment
D05 Perpetuals/ leveraged derivatives	Include leveraged derivative positions, especially perpetual claims with margin, liquidation and funding.	Perpetual is an instrument/ lifecycle subtype; "leveraged" is a margin property. Hyperliquid separately specifies margin, funding, liquidation and oracle-price behavior. [18]	**Merge into derivatives instrument family; preserve perp subtype.**
D06 Yield vaults/ aggregators/ managed strategies	Include pooled or wrapped capital deployed according to a strategy/mandate with share accounting.	"Yield" says nothing about mechanism. A deterministic wrapper, a strategy router and a curator-controlled allocator have different authority and risk. Morpho Vault V2 expressly separates owner, curator, allocator and sentinel authority. [19]	**Rename and split:** rule-based vaults vs delegated/curated asset management.
D07 Bridges/ cross-domain transfer	Include mechanisms/ services that verify and effect state or asset movement across settlement domains.	Infrastructure, not a peer financial product. LayerZero separates endpoint messaging, DVNs and executors; verification configuration determines trust. [20]	**Convert to settlement/ infrastructure taxonomy.**
D08 Intents/ aggregation/ order flow	Include architectures where a user specifies an outcome/order and routers, relayers or solvers choose execution.	Cross-cutting execution architecture. Across documents intents for transfer, transfer-plus-execution and swap-plus-execution. [21]	**Convert to mechanism/ execution facet.**
D09 RWA/ tokenized claims	Include tokens whose economic value or enforceability refers materially to an off-chain/pre-existing asset or legal claim.	This is primarily **reference-asset and legal-dependence information**, not one mechanism. FSB distinguishes native tokens from representations of pre-existing assets; Ondo's OUSG represents an LP interest and USDY is a tokenized note with specified backing. [22]	**Convert to reference-asset/ legal facets and actual instrument type.**

Existing area	Defensible operational rule	Principal defect	Recommended treatment
D10 Options/ structured products	Include options and other nonlinear/ packaged contingent payoffs.	Options are derivative instruments; “structured product” may instead be a managed vault holding derivatives.	**Merge options under derivatives; classify structured wrappers separately by product.**
D11 Reserve-backed/fiat stablecoins	Include peg-targeting claims whose redemption/backing depends materially on issuer-controlled reserves.	This describes backing, custody and legal trust. Circle reports dollar-denominated reserves with third-party custody/management and monthly attestations. [23]	**Convert to monetary-claim subtype plus custody/legal facets.**
D12 Prediction markets and other	Prediction subset: collateralized claims paying according to a future event resolution.	The prediction subset is coherent; “other” has no necessary condition at all. Polymarket’s outcome tokens are collateralized contingent claims with oracle-mediated resolution. [24]	**Preserve event-contingent subtype; remove “other.”**

The stablecoin boundary is particularly decisive. A two-way D03/D11 split cannot cover the current design space cleanly. Sky combines collateral-backed issuance and liquidation; Circle depends on issuer-held reserve assets and redemption; Ethena’s documentation describes USDe backing coupled to off-exchange custody and delta-neutral hedge mechanisms. Those systems all issue dollar-targeting claims but have radically different solvency, custody, derivatives, legal and oracle dependencies. [25]

Likewise, liquid staking and restaking deserve a parent-child relation rather than forced equivalence. Lido describes stETH as representing withdrawable underlying ETH, with rebases reflecting rewards or slashing; EigenLayer instead creates opt-in operator sets whose allocated stake can become slashable according to an AVS’s conditions. The latter introduces a security-service counterparty and slashing-authority graph that ordinary LST issuance does not predict. [26]

[Reproduced measurement on a proxy; medium confidence] Because the required DeFiFormal element matrix is unavailable, I performed a deliberately limited *category-level structural proxy* rather than pretending to run E02. Each D01–D12 label was encoded against 45 public-architecture attributes such as exchange, collateral, debt, issue/redeem, liquidation, derivative payoff, delegation/slashing, conditional payout, legal claim, custody, cross-domain settlement, relaying and allocation. Pairwise Jaccard overlap was then binned as High $\geq$0.30, Medium 0.14–0.30, Low <0.14:

	D01	D02	D03	D04	D05	D06	D07	D08	D09	D10	D11	D12
D01	—	L	L	L	**H**	L	L	L	L	**H**	L	**H**
D02	L	—	**H**	L	M	M	L	L	L	M	L	M

	D01	D02	D03	D04	D05	D06	D07	D08	D09	D10	D11	D12
D03	L	**H**	—	L	M	L	L	L	L	M	M	M
D04	L	L	L	—	L	L	L	L	L	L	L	M
D05	**H**	M	M	L	—	L	L	L	L	**H**	L	**H**
D06	L	M	L	L	L	—	L	L	L	L	L	L
D07	L	L	L	L	L	L	—	**H**	L	L	L	L
D08	L	L	L	L	L	L	**H**	—	L	L	L	L
D09	L	L	L	L	L	L	L	L	—	L	**H**	L
D10	**H**	M	M	L	**H**	L	L	L	L	—	L	**H**
D11	L	L	M	L	L	L	L	L	**H**	L	—	L
D12	**H**	M	M	M	**H**	L	L	L	L	**H**	L	—

Average-linkage hierarchical clustering of that proxy naturally produced, at six clusters, `{D01,D05,D10,D12}`, `{D02,D03}`, `{D07,D08}`, `{D09,D11}`, `{D04}`, and `{D06}`. The nearest-pair ordering was stable under Jaccard, cosine and Hamming distances: D10–D12, D09–D11 and D02–D03 remained the three closest pairs. **This is not E02/E04**: the inputs were researcher-coded category semantics, not DeFiFormal's 72 protocol decompositions. It is useful only as a structural sanity check.

The twelve quality tests therefore evaluate as follows:

Test	Finding
T01 Exhaustiveness	**Fail.** The explicit "other" bucket proves non-exhaustiveness by construction, and current analytics expose material omitted functions.
T02 Internal coherence	**Partial/fail.** D02 is relatively coherent; D12 is not; D06 mixes several management/control structures.
T03 Level consistency	**Fail strongly.** Products, mechanisms, instruments, infrastructure and legal/reference-asset concepts coexist at the same level.
T04 Boundary clarity	**Fail.** D02/D03, D05/D10, D07/D08, D09/D11 and D06 with several underlying areas are inherently overlapping.
T05 Orthogonality	**Fail.** The proxy analysis and primary architectures show systematic overlap rather than accidental edge cases.
T06 Empirical coverage	**Not fully reproducible.** External evidence demonstrates omitted material sectors, but corpus-level TVL/volume/OI/user coverage cannot be computed without the 72 roster.
T07 Temporal stability	**Weak.** Flat labels follow successive market waves; 2021, 2023 and 2026 descriptions emphasize different families. [27]

Test	Finding
T08 Cross-chain validity	**Partial/unvalidated.** Cross-domain facets are chain-agnostic, but the requested Ethereum/Solana/Cosmos/Bitcoin/Cardano/Midnight held-out experiment was not reproduced.
T09 Risk discrimination	**Fail.** Stablecoin, staking, bridge and vault examples show materially different risks inside or across nominal areas.
T10 Formal usefulness	**Partial.** Credit and derivative labels predict some state; RWA, intents and “yield” predict much less.
T11 Implementation usefulness	**Partial/fail.** Product labels do not reliably predict reusable semantic kernels.
T12 Inter-rater reproducibility	**Unresolved.** No independent reviewers were available; no κ/α statistic is reported or fabricated.

The strongest case for preserving the twelve, as required by the red-team brief, is nonetheless nontrivial. The set is compact, recognizable to DeFi practitioners, includes many economically important contemporary sectors, and—by the assignment’s description—supports a balanced five-applications-per-area construction benchmark. Current DefiLlama still uses labels closely resembling DEXs, Lending, Liquid Staking, Restaking, RWA, CDP, Yield, Derivatives and Prediction Markets, so the vocabulary is not obsolete in a market-navigation sense. [28] The twelve also impose low migration cost on already written briefs and constructions. **That is a strong argument for retaining the old identifiers as presentation tags and benchmark strata. It is not a strong argument for retaining them as the semantic ontology.**

External market evidence, missing areas, and validation

[Source fact; moving snapshot] DefiLlama’s current category page exposed roughly one hundred category labels at the research date. Among the categories visible in the current snapshot were Bridge, Liquid Staking, Lending, RWA, Staking Pool, DEXs, Restaking, Canonical Bridge, Onchain Capital Allocator, Risk Curators, CDP, Basis Trading, Yield, Derivatives, Liquid Restaking, Yield Aggregator, RWA Lending, Privacy and Prediction Market. Current displayed TVL figures included approximately $13.9 billion for Staking Pool, $9.34 billion for Onchain Capital Allocator, $9.26 billion for Risk Curators, $7.22 billion for Basis Trading and $0.88 billion for Privacy. These are observations of DefiLlama’s classification and methodology, **not additive market shares and not an ontology to copy**. [7]

The result is especially damaging to D06/D12 completeness. Current DefiLlama gives **Onchain Capital Allocator** and **Risk Curators** their own categories; the former describes pools actively controlled or managed by an operator or governance process, while the latter describes systems concerned with assessing and configuring DeFi risk. Morpho’s own Vault V2 architecture supports the substantive distinction: the curator configures adapters, caps, fees and other constraints, while allocator authority controls deployment within those boundaries. [29]

[Inference] A new flat “Risk Curator” category would nevertheless repeat the same mistake. Curation is best represented as **asset-management function + delegated-control mechanism + governance/**

authorization facets. A Morpho credit market remains credit; a vault allocating across those markets is an asset-management product; the curator is a control role. Those are distinct units.

Other omitted families are real rather than theoretical. DefiLlama currently exposes dedicated categories for payments, insurance, synthetics, privacy, options vaults, staking pools and launch mechanisms. Its payments page includes streaming/payment systems such as Superfluid and Sablier; its insurance page includes Nexus Mutual; its synthetics page includes Alchemix; and its options-vault category includes Rysk. [30]

A second major analytics provider independently reaches similar high-level functions. Messari's current classification system includes decentralized insurance, lending, liquid staking, payments and yield services. Token Terminal's sector methodology separately discusses exchanges, derivatives, lending, stablecoin issuers, liquid staking and asset management, and explicitly varies the economic metric by market sector. [31] That convergence is good evidence for broad **economic-function families**, but the disagreement and proliferation below that level are arguments against copying any provider's leaf labels wholesale.

[Source fact] TVL itself should not determine taxonomy. DefiLlama defines TVL in terms of the value of tokens locked in protocol/platform contracts; Token Terminal likewise chooses different economically meaningful metrics for different sectors. TVL is therefore useful as one empirical coverage coordinate, but it does not answer whether two applications have the same payoff, state machine, trust model or legal claim. [32]

Several external cases directly falsify top-level boundaries in the current twelve:

- **Stable-value claims:** Circle's USDC is backed by issuer-controlled dollar-denominated reserve assets with third-party custody/management and monthly attestation; Sky's USDS operates through protocol collateral and liquidation/peg mechanisms; Ethena's USDe uses backing assets plus hedging arrangements. Calling the first D11 and second D03 still leaves the third awkwardly between stablecoin, derivatives and managed strategy. [33]
- **Tokenized off-chain claims:** FSB explicitly separates native tokenized assets from representations of physical or pre-existing financial assets. Ondo documents OUSG as a limited-partnership interest and USDY as a tokenized note whose backing may include Treasury securities, an ETF or bank deposits. The decisive facts are instrument type, issuer/custodian, redemption right and legal enforceability—not "RWA mechanism." [22]
- **Intents and bridges:** Across specifies an intent/RFQ/relayer architecture that can support pure transfer and transfer-plus-execution; LayerZero decomposes interoperability into endpoint, verification networks and executors. These are execution/settlement architectures reusable underneath many economic functions. [34]
- **Event markets:** Polymarket's conditional-token architecture creates collateralized YES/NO claims, and resolution is supplied through an oracle/dispute process. Formally, this is much closer to a collateralized contingent claim with conditional settlement than to an amorphous "other" protocol. [35]
- **Restaking:** EigenLayer combines off-chain operator work with on-chain registration, rewards, allocation and slashing. The AVS—not merely a generic staking pool—can define service/slashing conditions. That is a distinct trust and obligation graph from ordinary liquid staking. [36]

Temporal audit. The evidence supports a useful pattern even though a fully archived 2021/2023/2026 protocol panel was not reconstructed. BIS's 2021 description centered on trading, lending and investing protocols plus stablecoins; a 2023 academic overview explicitly named exchanges, lending, derivatives and insurance; 2024 work described liquid staking and restaking as emerging innovations; by the September 2026 DefiLlama snapshot, Restaking, Liquid Restaking, Onchain Capital Allocator and Risk Curators all appear as explicit categories. [2] **[Inference]** This is exactly the failure mode expected from a flat zeitgeist taxonomy: new mechanisms or business models force new sibling categories even when the underlying economic functions and formal behaviors can already be expressed as combinations of stable facets.

Single-reviewer stress test, not E06. The following 24 cases illustrate how the linked model absorbs protocols that strain a flat label. They are **not** presented as a held-out inter-rater experiment; several sources helped design the taxonomy, and there was no independent reviewer pool. Their value is boundary testing only. Primary/current provider evidence for the cases comes from the cited protocol docs and analytics category records. [37]

Organization/ product	Linked assignment	What a flat label loses
Uniswap-style market	Exchange + spot + AMM	AMM is mechanism, not function
Aave	Credit + debt + pool/collateral/liquidation	Oracle/risk configuration differs by market
Morpho markets	Credit + debt + isolated-market mechanisms	Organization also contains asset-management vaults
Morpho Vaults	Asset management + share + curator/allocator	"Yield aggregator" hides delegated control
Sky/USDS	Credit **and** money issuance + debt/stable claim	Same system spans D02/D03/ D06-style functions
Lido/stETH	Staking + staking claim + delegation/rebase	Receipt accounting, validators and slashing are central
EigenLayer	Staking/security + slashable security commitment	Restaking adds AVS/operator obligations
Hyperliquid perps	Exchange + perpetual derivative + margin/ funding	"Leverage" is not the product ontology
Pendle	Exchange + maturity/yield claims + AMM	"Yield" can be an instrument being traded
Across	Cross-domain infrastructure + intent/relayer mechanism	D07 and D08 describe two axes of one system
LayerZero	Cross-domain messaging/verification infrastructure	Not a financial product at all

Organization/ product	Linked assignment	What a flat label loses
Ondo OUSG/ USDY	Asset management/issuance + fund/debt claim + legal/reference facets	“RWA” hides actual instrument and legal rights
Circle/USDC	Money issuance + reserve-backed monetary claim + custody/legal facets	Backing trust is more important than “fiat” label
Ethena/USDe	Money issuance + synthetic/hedged monetary claim + derivatives/custody facets	Does not fit D03/D11 dichotomy
Polymarket	Exchange + event-contingent claim + resolution oracle	Prediction is coherent; “other” is not
Nexus Mutual	Risk transfer + coverage claim	Missing economic function
Superfluid	Payments/streaming + time-indexed transfer mechanism	Missing economic function
Sablier	Payments/streaming/vesting	Missing economic function
SSV Network	Staking/security + validator/operator infrastructure	Staking service ≠ liquid staking receipt
Obol	Staking/security + validator/operator infrastructure	Same issue
Spark Liquidity Layer	Asset management/capital allocation	Delegated allocation is not adequately “yield”
Grove Finance	Asset management/capital allocation + external-asset/legal facets where applicable	Asset type and portfolio mandate are separate axes
Rysk option vault	Asset management + option/structured instrument	D06 and D10 overlap
Alchemix	Credit + synthetic/claim mechanics	Lending and synthetic claim labels overlap

The important result is not “24/24 classified,” because a sufficiently loose ontology can always do that. It is that **no `other` bucket is needed and the disputed cases become explicit multi-axis statements rather than arbitrary winner-take-all assignments.**

Recommended linked taxonomy and assignment rules

The recommended ontology treats **application/product** as the primary empirical unit. Protocol organizations are decomposed into products; products may contain markets/pools; their claims are separately typed as instruments; mechanisms and infrastructure are attached as multi-valued attributes.

Product families

Stable ID	Classified unit and defining question	Inclusion / exclusion	Examples and boundary counterexample	Formal and engineering utility
PF-EXCH — Exchange & market access	Application/ market. *Does it enable parties to enter, exit or exchange positions through a price-discovery/ matching process?*	Include spot AMMs, order books, RFQs, auctions and markets for contingent positions. Exclude routing-only and pure cross-domain messaging.	Uniswap-style spot market; Hyperliquid-style perp market; Polymarket-style event market. Across routing alone is not PF-EXCH. [38]	Predicts quotes/ orders, execution constraints, price formation and atomic settlement. Mechanism remains separately AMM/ book/RFQ.
PF-CREDIT — Credit & financing	Application/ market. *Does supplied capital create a repayable obligation or financed position?*	Include pooled/ isolated lending, secured/ unsecured credit and collateralized borrowing. Exclude staking rewards and non-repayable investment shares.	Aave/Morpho markets; Sky collateralized borrowing. A stablecoin redemption facility alone is not credit. [39]	Predicts principal, accrued debt, collateral, health, repayment, liquidation and bad debt.
PF-AM — Asset & portfolio management	Product. *Is capital pooled/ delegated under a strategy, mandate or curator-controlled envelope?*	Include vaults, index/fund wrappers, automated strategy routers and delegated allocators. Exclude a bare lending pool and mere UI routing.	Morpho Vault-style curator; rule-based vault; tokenized fund. Aave pool itself is PF-CREDIT, not automatically PF-AM. [40]	Predicts shares/ NAV, allocation state, caps, adapters, rebalancing authority and redemption constraints.

Stable ID	Classified unit and defining question	Inclusion / exclusion	Examples and boundary counterexample	Formal and engineering utility
PF-STAKE — Cryptoeconomic security & staking	Application/ service. *Is capital delegated or committed to secure consensus or another verifiable service for rewards and possible penalties?*	Include staking pools, LSTs, restaking and security delegation. Exclude a vault merely holding LSTs.	Lido; EigenLayer. A Yearn-like vault over stETH remains PF-AM with an IN-STAKE underlying dependency. [41]	Predicts stake, operator, reward, exit-delay and slash-state semantics.
PF-MONEY — Monetary claim issuance & redemption	Application/ issuer system. *Does it issue/ redeem a claim targeting a monetary unit or reference value?*	Include reserve-backed, collateral-backed, synthetic/ hedged and hybrid stable-value issuance. Exclude ordinary fund shares and asset wrappers without a monetary target.	USDC, USDS, USDe. [42]	Predicts mint/ burn/ redemption, backing/ solvency, peg-support interface and reserve/ oracle dependencies.
PF-PAY — Payments, escrow & streaming	Payment arrangement. *Is the principal function conditional or scheduled value transfer rather than market-position creation?*	Include escrow, payment channels, streaming, vesting and disbursement. Exclude secondary-market trading and cross-chain verification itself.	Superfluid/Sablier-style streaming. DefiLlama currently recognizes a Payments category. [43]	Highly reusable bounded state machines: deposits, release, cancellation, schedule and timeout.

Stable ID	Classified unit and defining question	Inclusion / exclusion	Examples and boundary counterexample	Formal and engineering utility
PF-RISK — Risk transfer & insurance	Coverage product. *Is a defined loss exposure transferred to another capital provider/pool?*	Include mutual cover and parametric/ adjudicated insurance. Exclude ordinary collateral liquidation and an issuer's internal hedge.	Nexus Mutual-style cover; DefiLlama and Messari both maintain insurance concepts. [44]	Predicts premiums, coverage limits, trigger/ adjudication, reserves and payout obligations.
PF-CAP — Capital formation & distribution	Issuance/ distribution program. *Does it allocate newly issued assets/ rights through a sale, auction, vesting or distribution rule?*	Include launch auctions, primary token sales and vesting/ distribution. Exclude routine stablecoin redemption and secondary trading.	Token launch auction/vesting distributor; current analytics maintain launchpad-style categories. [45]	Predicts supply caps, eligibility, allocation, refunds and vesting schedules.

These product families deliberately **do not contain "derivatives," "RWA," "AMM," "intents," or "bridges" as peer families**. Those concepts answer different questions.

Instrument and payoff families

The instrument taxonomy should use a separate namespace:

ID	Instrument/payoff family	Important subtypes
IN-SPOT	Immediate-delivery asset position	fungible token, LP asset, commodity-like token
IN-DEBT	Debt/receivable	secured/unsecured; fixed/floating; fixed/open maturity
IN-SHARE	Pooled/fund share	vault share, LP share, index/fund interest
IN-MONEY	Peg-targeting monetary claim	reserve-backed; collateralized debt; synthetic/hedged; algorithmic/hybrid
IN-STAKE	Staking/security claim	LST, liquid-restaking claim, non-transferable stake
IN-DERIV	Derivative/contingent claim	perpetual; future/forward; option; swap; event-contingent; structured/yield claim

ID	Instrument/payoff family	Important subtypes
IN-COVER	Insurance/coverage claim	parametric, adjudicated, mutual

[Recommendation] D05 and D10 therefore merge only at this **instrument-family level**: both are `IN-DERIV`, while `perpetual` and `option` remain strongly distinct subtypes because their lifecycle, exercise, margin and payoff equations differ. Current derivatives research likewise treats perpetuals, options and synthetics as distinguishable instrument classes while seeking a common protocol-agnostic framework. [46]

Prediction markets become `PF-EXCH + IN-DERIV:event-contingent + ME-COND`, not an "other" market. Polymarket's current architecture demonstrates precisely the formal signature: collateral splits into outcome claims, claims are traded, an external resolution determines the winning payoff, and winning claims redeem against collateral. [47]

Mechanism taxonomy

Mechanisms are multi-label and reusable:

`ME-POOL` pooled accounting/pro-rata shares; `ME-AMM` curve-based pricing; `ME-BOOK` order book; `ME-RFQ` RFQ/auction; `ME-DEBT` debt accrual; `ME-COLL` collateral health/liquidation; `ME-ISSUE` issue/burn/redeem; `ME-MARGIN` margin/mark-to-market/funding; `ME-COND` conditional resolution/payout; `ME-ALLOC` allocation/caps/rebalancing; `ME-STAKE` delegation/reward/exit/slashing; `ME-INTENT` intent/solver/router execution; `ME-XDOM` cross-domain settlement; `ME-ORACLE` oracle/attestation consumption; and `ME-AUTO` keeper/automation.

This separation fixes several current contradictions. A spot exchange may be `PF-EXCH + IN-SPOT + ME-AMM`, or equally `PF-EXCH + IN-SPOT + ME-BOOK`; a perp venue may use the same order book but add `IN-DERIV:perpetual + ME-MARGIN`; an intent-driven spot swap adds `ME-INTENT` without becoming a new economic product. Across's documentation is direct evidence that intent execution spans multiple outcomes, while LayerZero shows that cross-domain verification and execution have their own infrastructure actors. [48]

Infrastructure and trust facets

The following should be mandatory profile fields rather than optional prose: settlement domain; custody/asset-control model; oracle/external-data model; authorization/identity; governance and upgrade authority; off-chain legal dependence; sequencing/ordering; validator/operator dependence; cross-domain verification; privacy/disclosure; composability dependencies; collateral/solvency model; liquidity model; maturity/lifecycle; issuance/redemption; price discovery; and principal failure modes.

The need is empirical. LayerZero recommends multiple independent verification networks because verification configuration affects forgery risk; Morpho exposes role- and timelock-based control over allocations and caps; Lido's receipt value can reflect validator penalties; Ondo token rights depend on legal entities; Circle depends on reserve custody; and EigenLayer's delegated stake can be exposed to AVS-defined slashing. A single marketing-category label does not discriminate any of those trust boundaries. [49]

The complete D01–D12 crosswalk is therefore:

Old	New mapping	Decision
D01	`PF-EXCH + IN-SPOT + {ME-AMM, ME-BOOK, ME-RFQ…}`	**Rename/narrow**
D02	`PF-CREDIT + IN-DEBT + ME-DEBT (+ ME-COLL where applicable)`	**Preserve/rename**
D03	`PF-CREDIT + PF-MONEY + IN-DEBT/IN-MONEY + ME-COLL + ME-ISSUE`	**Merge/convert**
D04	`PF-STAKE + IN-STAKE + ME-STAKE`; subtype LST/restaking	**Split under parent**
D05	`PF-EXCH + IN-DERIV:perpetual + ME-MARGIN`	**Merge under derivatives instrument family**
D06	`PF-AM + IN-SHARE + ME-ALLOC`; control facet identifies curator/discretion	**Rename/split**
D07	`ME-XDOM + settlement/verification infrastructure facets`	**Convert to infrastructure**
D08	`ME-INTENT + execution/order-flow facets`	**Convert to mechanism**
D09	actual `PF-*` and `IN-*` + `reference_asset=offchain` + legal/custody facets	**Convert to facets**
D10	`PF-EXCH` or `PF-AM` + `IN-DERIV:option/structured`	**Merge/split by actual product**
D11	`PF-MONEY + IN-MONEY:reserve-backed + custody/legal facets`	**Convert to subtype/facets**
D12	prediction → `PF-EXCH + IN-DERIV:event + ME-COND`; "other" dissolved	**Split/remove "other"**

[Unresolved] The requested corresponding **72-row protocol mapping cannot be produced** because the 72 identities and versions have not been independently recovered. The downloadable crosswalk intentionally contains zero fabricated corpus assignments.

Decision scorecard and Moriarty design

The raw scoring below follows the assignment's weights. Scores are **[Recommendation/inference]**, not falsely presented as corpus measurements. M0 is the current flat twelve; M1 a repaired flat list; M2 a hierarchy; M3 faceted multi-label; M4 linked product/mechanism classifications; M5 formal-behavior-only.

Model	Exhaust. 18%	Coher. 14%	Boundary 12%	Mechanism/ risk 12%	Formal 14%	Moriarty 12%	Temporal/ cross-chain 8%	Migration 5%	Maintain. 5%
M0	1.8	2.0	1.4	1.5	1.8	1.5	1.6	**5.0**	2.8
M1	3.3	3.1	2.7	2.7	2.8	2.6	2.8	4.2	3.3
M2	4.1	4.0	3.8	3.5	3.7	3.5	3.8	3.4	4.1
M3	4.8	4.2	4.6	4.6	4.5	4.3	**4.8**	2.5	4.0
M4	**4.9**	**4.7**	**4.7**	4.9	4.9	4.9	**4.8**	3.2	**4.5**
M5	3.8	**4.7**	4.0	**5.0**	**5.0**	**5.0**	**4.8**	1.8	3.4

[Reproduced measurement on scoring assumptions] A 100,000-draw sensitivity exercise varied the requested criterion weights around their baseline using a Dirichlet distribution and independently perturbed each raw score by a standard deviation of 0.25 points. Under those assumptions M4 ranked first in **99.19%** of draws, M3 in 0.67%, and M5 in 0.14%. This demonstrates robustness **conditional on the judgment scores**, not external statistical proof of taxonomy superiority.

Another model can still win under explicit preference changes. M5 becomes attractive if the taxonomy exists almost exclusively for formal verification and Moriarty implementation and market-facing explanatory value is heavily downweighted. M3 can win if one values a single flexible annotation system more than a clean product/mechanism separation. M2 becomes credible where tooling insists on one navigable tree. M0's only material advantage is migration: it would become rational only if preservation of existing documents overwhelmingly dominated exhaustiveness, risk discrimination and semantic reuse.

Moriarty should compile behaviors, not market labels

The Marlowe precedent is important. Marlowe defines executable semantics around transaction input, contract state and contract continuation; its test documentation describes the contract as a continuation DAG whose branching arises from inputs and timeouts. Its formal guarantees attach to that behavior, not to whether the contract is marketed as a "loan," "swap," or "derivative." The same documentation also shows why implementation constraints must remain distinct from abstract semantics: Cardano protocol limits can block transactions even when abstract Marlowe semantics permit them. [14]

[Recommendation] Moriarty should therefore have a small semantic kernel built around reusable obligations and transitions:

- party/role and authorization;
- assets, balances, escrow and transfer;
- issue/burn/redeem;
- pooled shares and proportional accounting;
- debt and accrual;
- collateral, solvency and liquidation;
- quote/order/auction/matching predicates;
- margin, funding and mark-to-market;

- schedules, maturity and timeout;
- conditional resolution/payout;
- allocation limits and delegated authority;
- oracle/attestation *consumption*;
- reward/penalty/slashing accounting.

Product-family names should be **library metadata and construction templates**, not language primitives.

Current Midnight tooling reinforces that architecture. Compact compilation produces executable contract artifacts, proving/verifying keys and ZKIR; Midnight describes ZKIR as the intermediate representation derived from contract circuits and the bridge from Compact logic to the zero-knowledge backend. Midnight.js then separates local circuit execution, proof generation, transaction balancing, broadcast and finalization. [50] **[Inference]** That compiler/runtime boundary is a natural place for Moriarty to keep deterministic financial semantics separate from oracle acquisition, solver discovery, cross-domain relaying, legal enforcement and other open-world processes.

Current Midnight documentation also makes maintenance/upgradability an explicit deployment concern because proving schemes, ZKIR formats or verifier-key formats may evolve. That is further evidence that **upgrade authority belongs in the trust/control profile**, rather than being silently inherited from a product label. [51]

A canonical Moriarty matrix for the recommended product families is:

Family	Canonical Moriarty application	Actors, state and actions	Core safety / liveness obligations	Compact/ZKIR and external boundary
PF-EXCH	Bounded swap/auction or matched order	maker/taker/LP; assets, quotes/orders, liquidity; place/cancel/fill/settle	asset conservation; min-received/limit-price constraint; no double fill; cancel/expiry liveness	Matching/AMM predicates can be libraries; private order facts can become Compact proofs. Global solver/order discovery stays Runtime.
PF-CREDIT	Collateralized loan	borrower, lender/pool, liquidator, oracle; collateral, debt, rate, health; borrow/repay/add/remove/liquidate	no unauthorized collateral release; debt accounting; liquidation conservation; terminal repayment/close paths	Health and authorized transition proofs fit Compact. Price acquisition, keeper competition and off-chain credit enforcement remain external.

Family	Canonical Moriarty application	Actors, state and actions	Core safety / liveness obligations	Compact/ZKIR and external boundary
PF-AM	Capped share vault with allocator	depositor, curator, allocator; shares, assets, adapter exposures/caps; deposit/redeem/rebalance	share/asset accounting; allocation caps; authority bounds; redeemability assumptions	Cap/authority proofs fit Compact; external adapters are explicit capabilities. Discretionary investment judgment stays outside Core.
PF-STAKE	Delegated stake receipt	staker, operator, security service; stake/shares/rewards/slash liability; delegate/exit/reward/slash	no over-redemption; penalties allocated according to rule; exit delay respected	Receipt/reward/slash accounting can be modeled; validator duties, consensus evidence and AVS fault adjudication are external capabilities. Lido/EigenLayer show why. [52]
PF-MONEY	Backed claim mint/redeem	minter/issuer/redeemer, collateral provider/custodian/oracle; supply/backing; mint/burn/redeem/liquidate	supply conservation; backing or collateral rules; authorized redemption	On-chain collateral constraints can be proven. A market-price peg cannot be guaranteed by DSL semantics; reserve custody, law and off-chain attestation stay external.
PF-PAY	Escrow/stream/vesting contract	payer, payee, optional arbitrator; locked amount, schedule; fund/claim/cancel/refund	no overpayment; schedule correctness; refund/timeout path; no permanently locked residual value	Excellent Moriarty-Core candidate; private eligibility can be proven in Compact.
PF-RISK	Parametric cover	policyholder, capital pool, resolver; premium, coverage, trigger, reserves; buy/resolve/pay	coverage cap; reserve/payout accounting; one settlement per covered event	Parametric payout semantics fit library/Core. Real-world loss adjudication/oracle protocol remains external.

Family	Canonical Moriarty application	Actors, state and actions	Core safety / liveness obligations	Compact/ZKIR and external boundary
PF-CAP	Primary sale + vesting	issuer, purchaser, beneficiary; sale cap, allocations, vest schedule; buy/refund/claim	supply cap; allocation conservation; refund after failure; vesting monotonicity	Eligibility/privacy predicates fit Compact; KYC/legal sale restrictions remain external policy/capabilities.

Derivatives should then be **Moriarty libraries parameterized by payoff/lifecycle**, not top-level core syntax. A perpetual library adds margin, funding and liquidation; an option library adds strike, expiry and exercise; an event-contract library adds a resolution value and conditional payout. Hyperliquid's current distinction among margin, funding, liquidation and oracle pricing and Polymarket's split/merge/redeem lifecycle illustrate that these are genuinely different state-machine extensions even though both sit under `IN-DERIV`. [53]

Keep out of Moriarty Core: solver search and competition, oracle data acquisition, bridge/DVN verification protocols, sequencer behavior, external custodian operations, validator operation, legal enforcement, KYC providers, discretionary portfolio decisions, governance politics, off-chain matching infrastructure and chain-specific deployment plumbing. These can be represented by typed capabilities, assumptions and proof obligations, but not pretended into deterministic financial semantics. LayerZero, Across, Ondo, Circle and EigenLayer each demonstrate why these open-world dependencies deserve first-class trust declarations. [54]

Final disposition

1. Preserve

[Recommendation] Preserve **D02's core concept** as **PF-CREDIT — Credit & Financing**. Preserve the coherent prediction-market portion of D12, but only as an event-contingent instrument/market subtype. Preserve "spot," "perpetual," "option," "liquid staking," "restaking," "stablecoin," "RWA," "bridge" and "intent" terms as useful human-facing labels, tags and benchmark strata—not as peers in one exclusive ontology.

2. Rename

[Recommendation] Rename D01's product concept to **Exchange & Market Access**, with AMM moved to mechanism. Rename D06's coherent part to **Asset & Portfolio Management**. Rename D04's common parent to **Cryptoeconomic Security & Staking**, underneath which liquid staking and restaking remain separate subtypes.

3. Merge

[Recommendation] Merge D05 and D10 at the **instrument-family level** into `IN-DERIV`, retaining perpetual, future/forward, option, event-contingent and structured-payoff subtypes. Merge the debt aspect of D03 into PF-CREDIT and its monetary-claim aspect into PF-MONEY rather than forcing a single exclusive home. Current derivatives research supports a common architecture with instrument-specific dynamics. [55]

4. Split

[Recommendation] Split D04 into liquid staking and restaking/security-service subtypes because their slashing and operator obligations differ. Split D06 into rule-based/automated vaults and delegated/curated asset management. Split D10's "options" from the broader category of structured wrappers: an option is an instrument; a vault holding options is an asset-management product. Split D12 permanently into coherent prediction/event markets and the individually reclassified former "other" members. [56]

5. Add

[Recommendation] Add economic-function families for **Payments/Escrow/Streaming**, **Risk Transfer/Insurance**, and **Capital Formation/Distribution**. Add **delegated/curated capital allocation** as a subtype of Asset & Portfolio Management rather than a disconnected top-level sector. Current DefiLlama and Messari classifications independently recognize payments and insurance, while current DefiLlama data separately surfaces large capital-allocator and risk-curator sectors. [57]

6. Convert to facets

[Recommendation] Convert **RWA** into reference-asset, legal-dependence, identity and custody facets; **reserve-backed/fiat stablecoin** into monetary-instrument/backing/custody/legal facets; **intents/aggregation/order flow** into execution architecture; **bridging** into settlement-domain/interoperability infrastructure; **AMM** into price-discovery/liquidity mechanism; and "leveraged" into collateral/margin characteristics. FSB's distinction between native and represented assets and the architecture of Across/LayerZero strongly support these level separations. [58]

7. Remove

[Recommendation] Remove **"other"** as a classification category. Remove "yield" as a defining mechanism: yield is an economic outcome generated by many unrelated mechanisms. Remove "RWA," "bridge," and "intent" from the exclusive top-level product partition. Do not remove those concepts from the ontology; move them to the correct typed dimensions.

8. Reclassify

[Unresolved question] The exact reclassification of all **72 alleged DeFiFormal protocols cannot responsibly be supplied from the evidence available on September 2, 2026**. The requested public repository is not retrievable, so the identities, versions and original memberships cannot be verified. [3] The machine-readable ontology deliberately leaves `corpus_protocol_assignments` empty rather than constructing a fictitious 72-row answer. Once a verifiable source commit exists, each application/product should be mapped to `PF-*`, each financial claim to `IN-*`, each mechanism to `ME-*`, and each dependency to the trust facets above; protocol organizations may then aggregate those child assignments without forcing a single category.

9. Implement in Moriarty

[Recommendation] Implement reusable primitives for asset/account conservation, transfer/escrow, issue/burn/redeem, pool/share accounting, debt accrual, collateral/solvency/liquidation, order/auction constraints, margin/funding, schedules/timeouts, contingent settlement, delegated allocation limits and reward/penalty accounting. Build canonical product libraries for PF-EXCH, PF-CREDIT, PF-AM, PF-STAKE, PF-MONEY, PF-PAY, PF-RISK and PF-CAP. Compile deterministic predicates and privacy-sensitive authorization/accounting into Compact and ultimately ZKIR where appropriate. Midnight's current toolchain explicitly compiles Compact contracts through ZK artifacts including ZKIR, supporting this layered design. [59]

10. Keep outside Moriarty Core

[Recommendation] Keep market-sector names, off-chain legal rights, external reserve custody, oracle acquisition, sequencer guarantees, solver search, relayer networks, bridge verification, validator operation, KYC/identity providers, discretionary investment judgment and governance politics outside the semantic kernel. Represent them as libraries, Compact modules, typed Runtime capabilities, oracle protocols, infrastructure adapters and explicit assumptions. This preserves what Marlowe-style DSLs do best—a bounded state-transition semantics with checkable obligations—while preventing Moriarty from falsely treating open-world institutional or infrastructure assumptions as on-chain financial logic. [60]

Final decision: replace the flat twelve-area ontology with M4+—four linked, multi-label classifications plus a formal-behavior profile. Keep the old D01–D12 identifiers only as legacy reporting tags. The decisive design principle is that **a DeFi product answers what financial service is being provided; an instrument answers what claim or payoff exists; a mechanism answers how it is implemented; and infrastructure/trust facets answer what the implementation must rely on. Moriarty should formalize the resulting behaviors and obligations, not the marketing category.**

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